QUANTUM SOURCE

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Date

A Compound Photon–Atom Architecture for High-Performance Fault-Tolerant Quantum Computing

Deterministic entanglement, nonlocal connectivity, and fast error-correction cycles

Quantum Source Labs – Confidential Information

Outline

1. **Why numbers alone are not enough for fault-tolerant quantum computing (FTQC)**

   Numbers, connectivity, and the limitations of current modalities

2. **Why photon–atom**

   Photonic connectivity + deterministic atom-mediated entanglement

3. **How the architecture scales**

   Unit cells, routing, RHG fault tolerance, and the hardware stack

4. **Where the performance edge comes from**

   Fast EC cycles, low routing overhead, efficient logical operations, and projected application performance

Fault tolerance is a problem of scale

Today's best physical qubits run at 10^{-3}–10^{-4} error rates. Useful algorithms need logical error rates below 10^{-12}. Closing that gap costs qubits — a lot of them.

10^{-3}–10^{-4}

Physical qubit / gate error rate today

$<10^{-12}$

Logical error rate needed for useful algorithms

10^{2}–10^{3}

Physical qubits per logical qubit at these rates

$\sim 10^{6}$

Physical qubits for a few-thousand-logical-qubit algorithm

Scale = Numbers + Connectivity

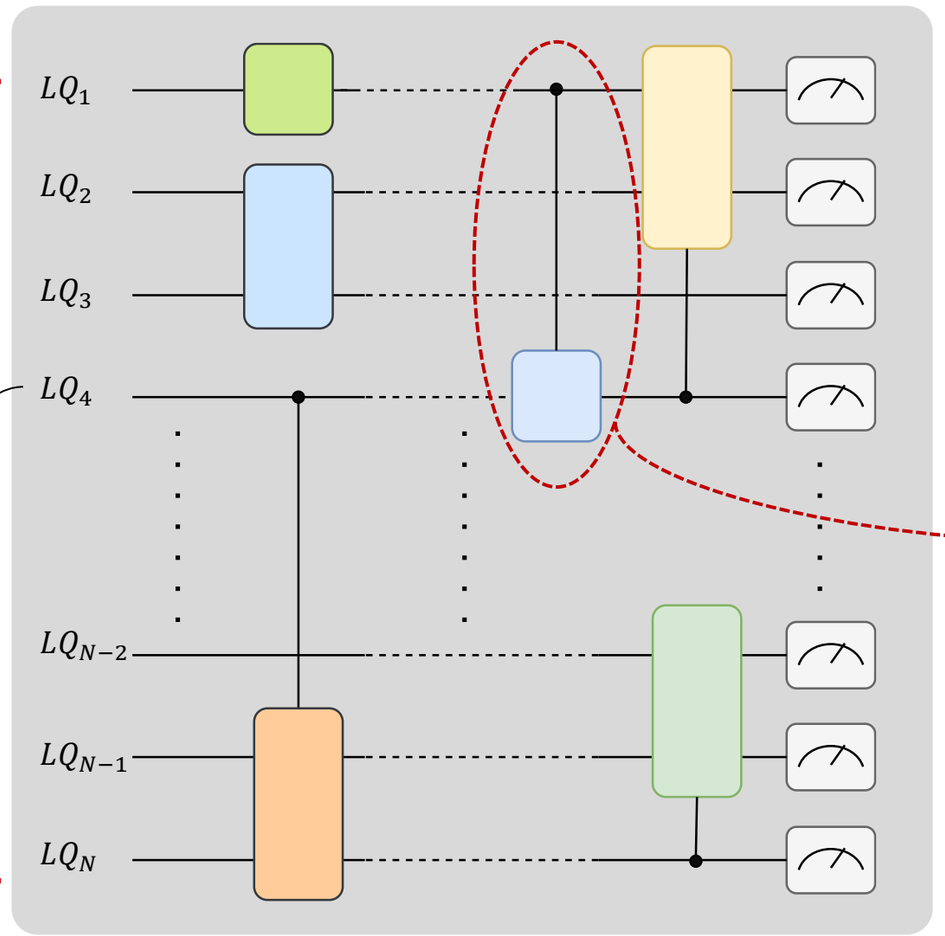
The stats above cover the numbers. The half that's easy to overlook: connectivity — entangling distant qubits efficiently matters just as much, and determines which compilations and error-correcting codes are even possible.

FTQC: what's under the hood?

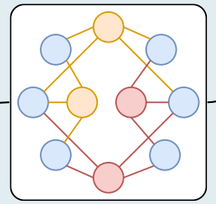
"Utility-scale QC" (Breaking RSA-2048, QChem, Finance,…)

Logical circuit depth > 10^{10} *operations*

$N > 1000$ *Logical qubits*

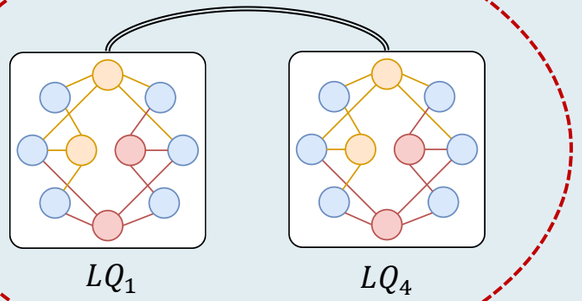


$n \sim 100 - 1000$ physical qubits

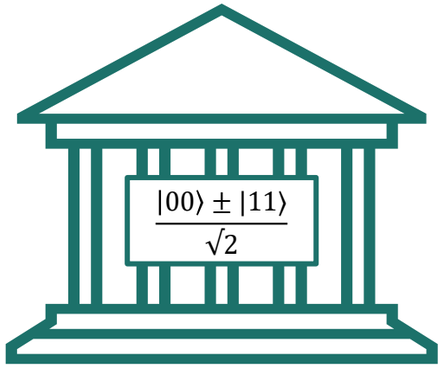


Logical qubit encoded into a Quantum Error Correction Code

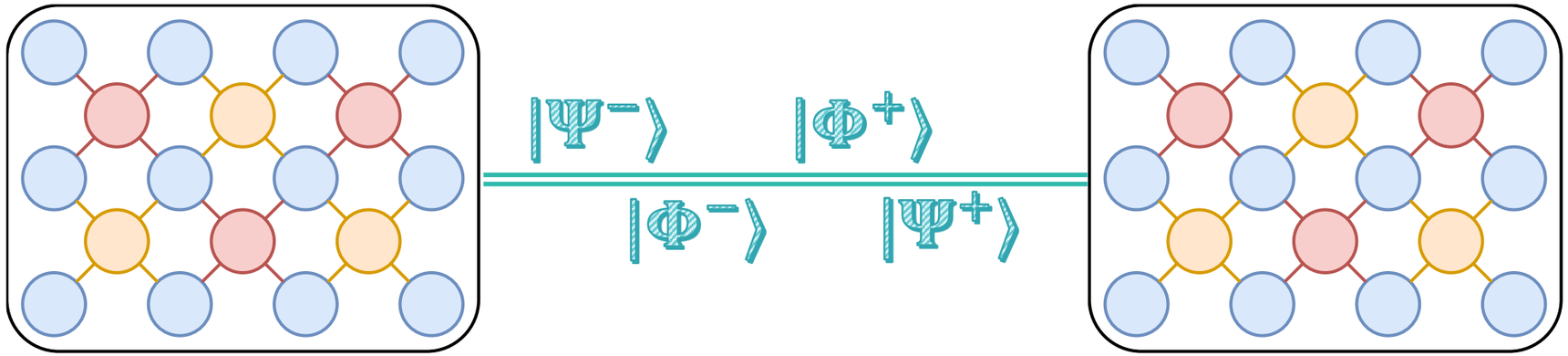
Logical two-qubit gate



Connectivity provides entanglement liquidity



- Entanglement is the currency of quantum computing
- Entanglement has to be distributed, not just generated.
- Every non-local interaction not supported natively gets taxed: in swaps, in shuttling, in fidelity.
- Modular scale-out just moves the connectivity problem to the module boundary.
- Logical gates need entanglement across all logical qubits, not just across neighbors.



Matter-based modalities: hurdles to connectivity



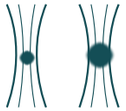
Superconducting

Planar chip layouts limit interactions largely to nearest neighbors; extending connectivity means added crosstalk and control complexity.



Trapped ion

Shared motional modes crowd as more ions are added, degrading two-qubit gate fidelity as the register grows.



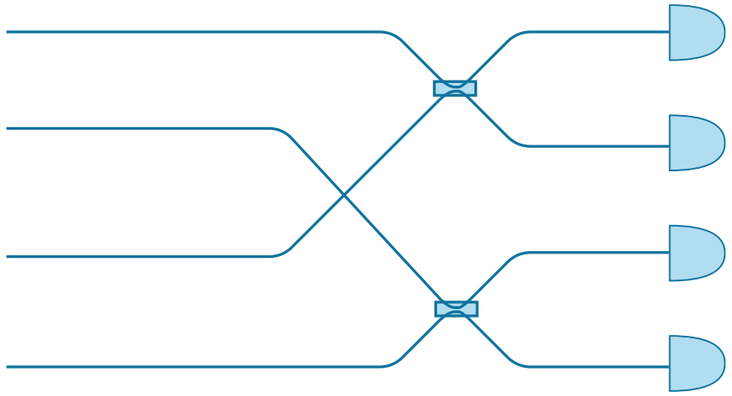
Neutral atom

Reconfigurable via shuttling, but mid-circuit moves and measurements push QEC rounds to milliseconds — 1000× slower than superconducting.

Modular scale-out (QPU-to-QPU links) is the common fallback — but reliable inter-module connections become the next hard problem.

Photons solve connectivity — and create a new problem

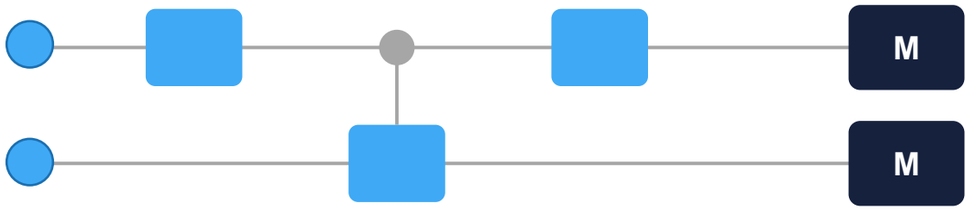
- In purely photonic architectures, photons both carry the logical quantum information and distribute the entanglement.
- Routing is nearly free: photons can be sent anywhere in the system at negligible marginal cost.
- Decoherence is negligible: photons barely interact with their environment, so the information they carry stays clean over long distances and long times.
- **The catch: generation and entanglement are probabilistic leading to massive resource overheads.**





Computing with photons: the case for MBQC

Circuit-based (CBQC, familiar)



Qubits are held in place; gates are applied to them over time; they're measured only at the end.

Measurement-based (MBQC)



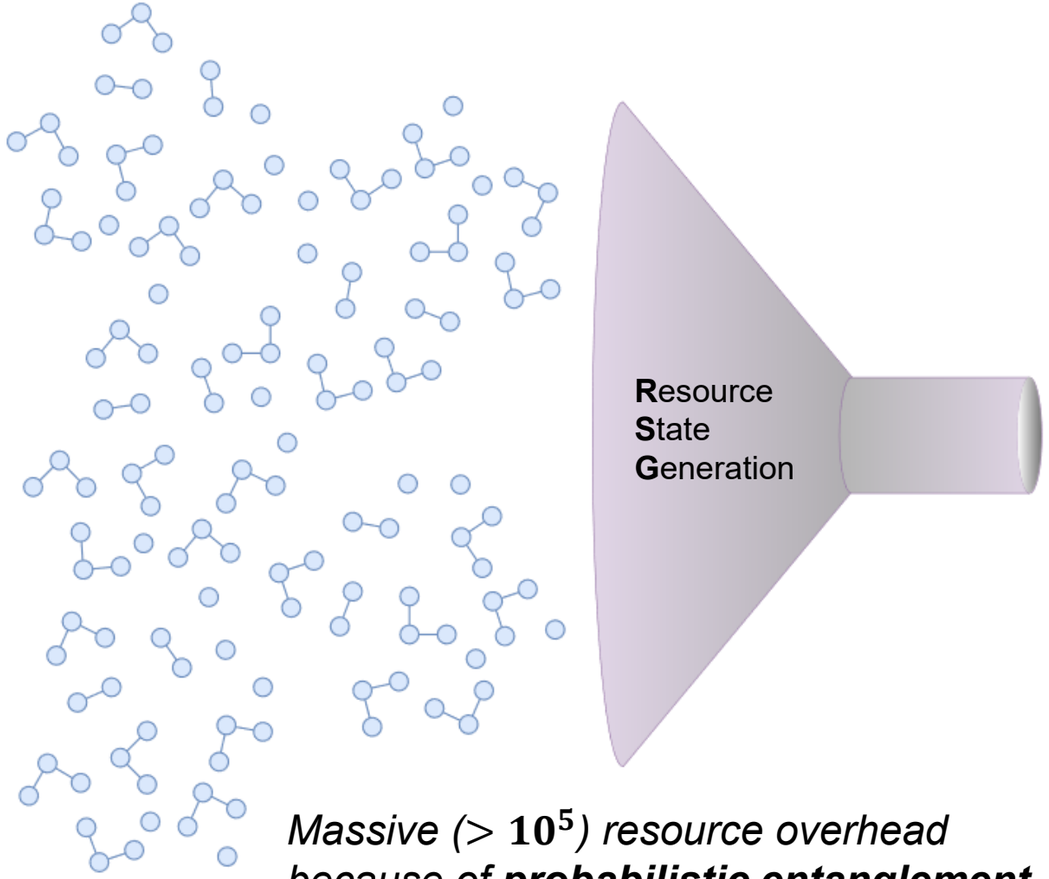
A large entangled **cluster state** is built up front. After that it's just single-qubit measurements — later measurement bases can depend on earlier outcomes (feed-forward), which is what implements the logic.

Why this is a natural fit for photons

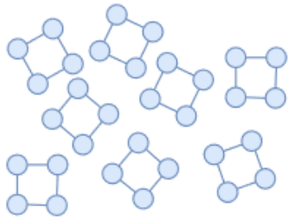
A photon is generated, entangled, measured, and gone — it was never going to sit still for a sequence of gates. MBQC only ever asks a qubit to be entangled a few times and then measured once, so a photon's one-shot lifecycle matches the computing model exactly. The circuit model, by contrast, expects a persistent qubit you can keep coming back to.

Photonic FTQC: scale through resource states generation

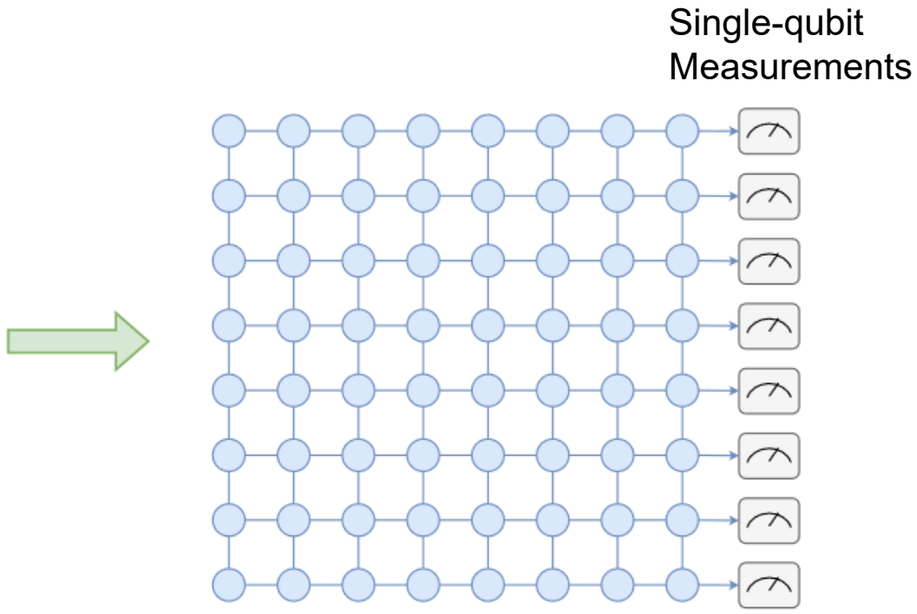
Rather than growing one qubit at a time, fully photonic architectures assemble the cluster state from small resource states generated in parallel and stitched together



*Massive ($> 10^5$) resource overhead because of **probabilistic entanglement***



Resource States



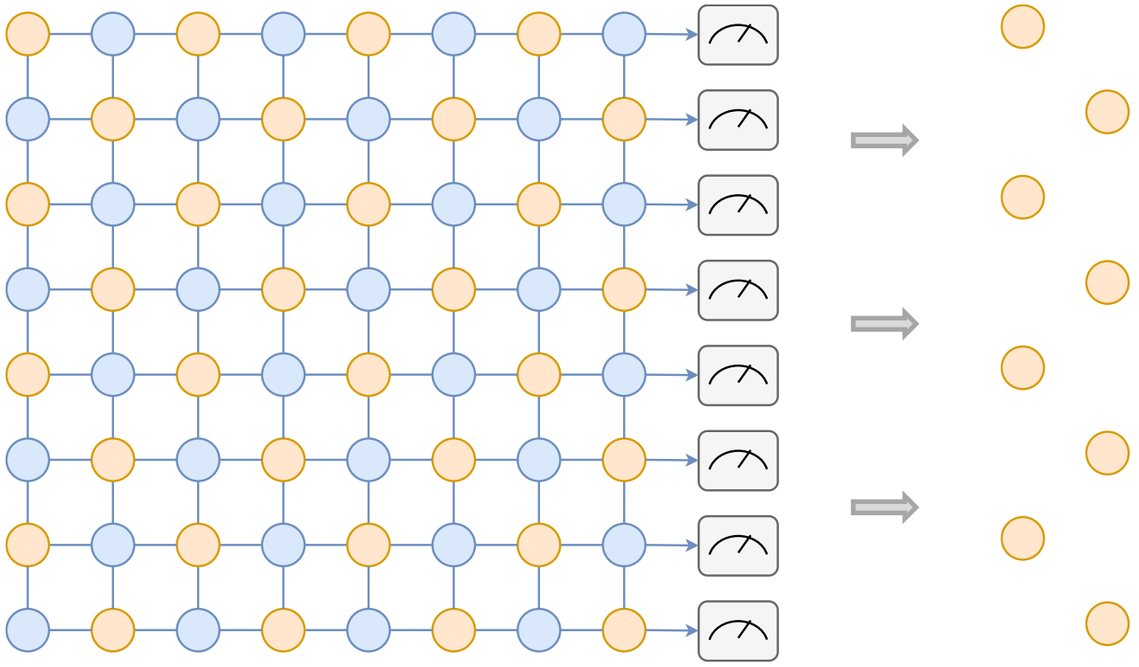
Photon Cluster State

Deterministic entanglement: taming the resource blowup

Let's assume two kinds of qubits that can entangle with each other, near-deterministically:

Photons Atoms

Photon-Atom Cluster State



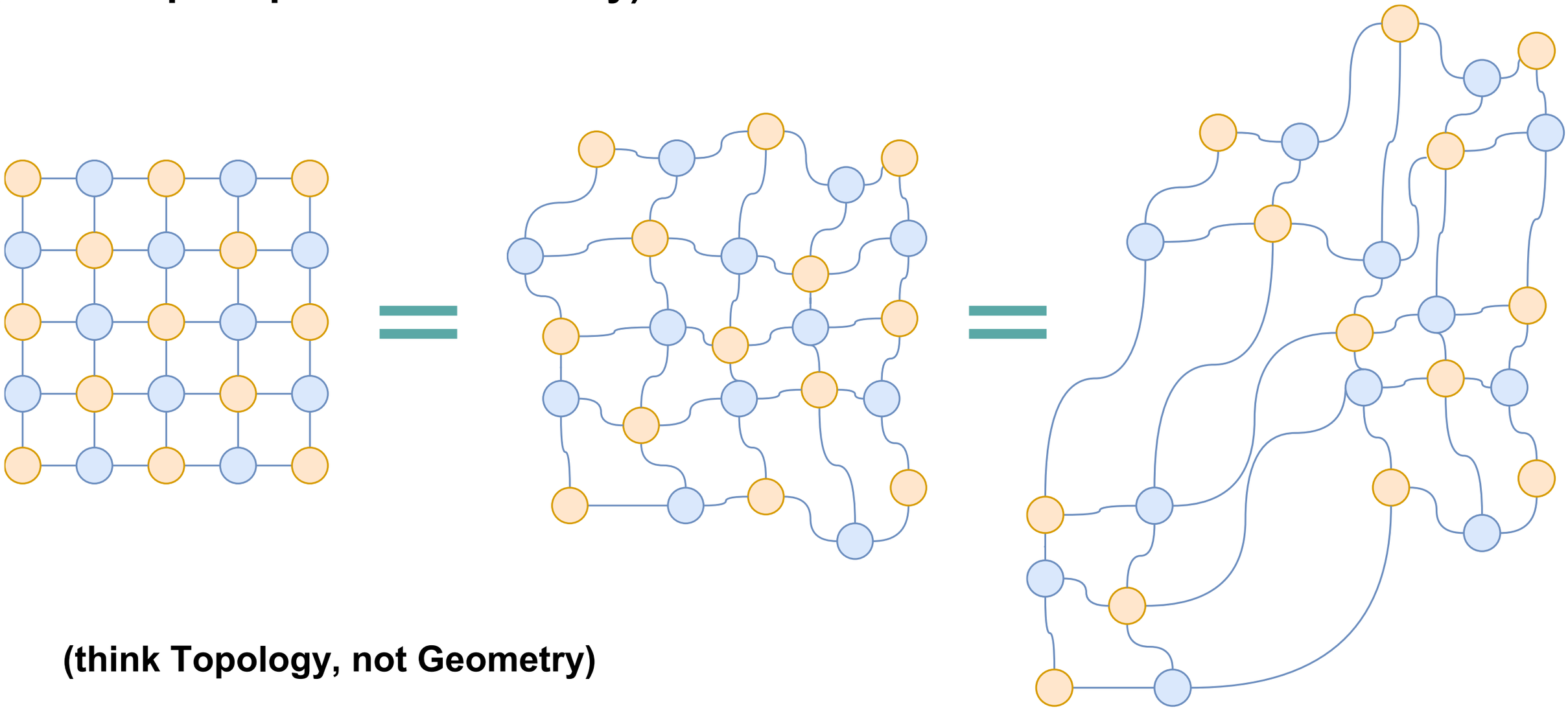
Measured atoms are recycled!

Measure a photon, and it's gone.

Measure an atom, and it's still there, ready to be reinitialized and reused.

Resource overhead: eliminated

Photon-atom cluster state preserves unlimited connectivity
(like in a pure photonic modality)



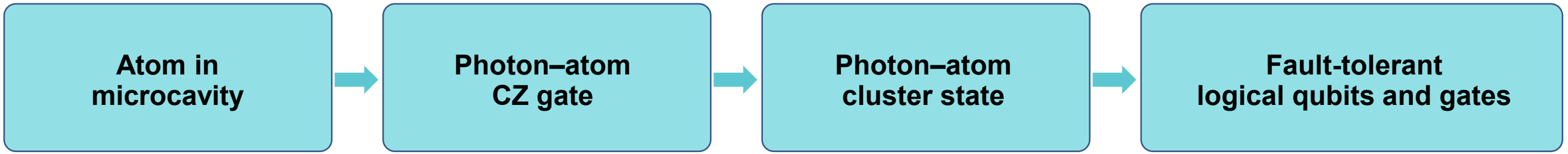
(think Topology, not Geometry)

Making it work: the compound photon-atom architecture

arXiv:2606.30385

Powered by Cavity QED

A single atom trapped inside a high-cooperativity cavity enables a **near-deterministic photon–atom gate**.



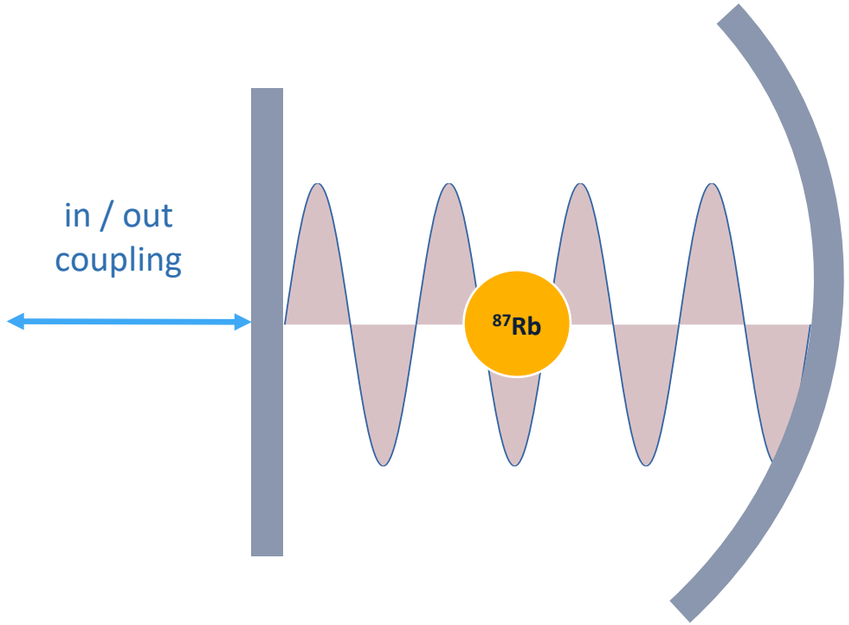
The core primitive: a controlled-phase gate between one atom and one dual-rail photon — robust to imperfections, implemented on ~10 ns timescales.

- Photons → long-range connectivity & scale
- Atoms → reusable, deterministic entangling resource

} Both photons and atoms carry quantum information

Cavity QED, in a nutshell

- A single atom placed inside a high-finesse optical cavity, coupled to one confined light mode
- Confinement lets the atom and cavity field exchange a photon at a high rate
- Turns a free-space atom–photon interaction (well under 1%) into near-unit-probability coupling — the deterministic gate depends on this
- One main parameter, the cooperativity $C \simeq 100$, sets the efficiency of the CZ gate, SPAM, and photon generation alike — a single dial tunes the whole platform



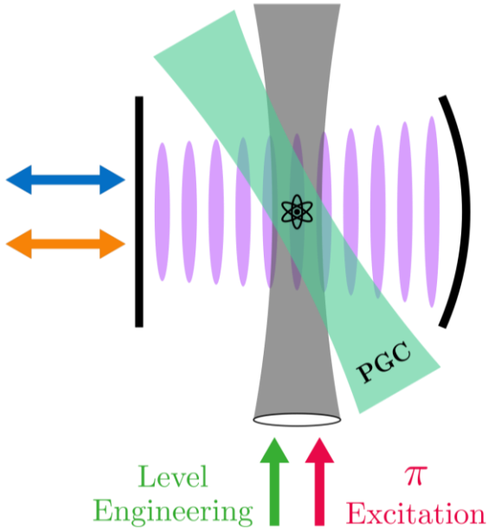
The unit cell: one atom, one cavity, every core operation

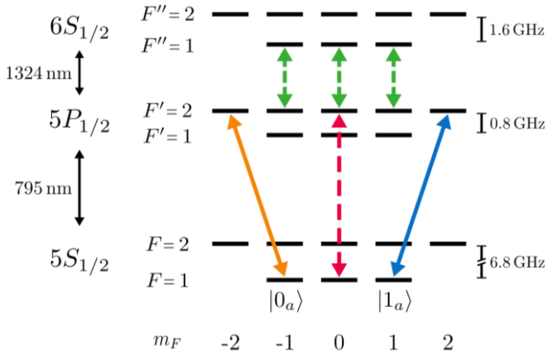
One reusable module for:

- ✓ entangling gates
- ✓ single-photon generation
- ✓ atomic state preparation / measurement

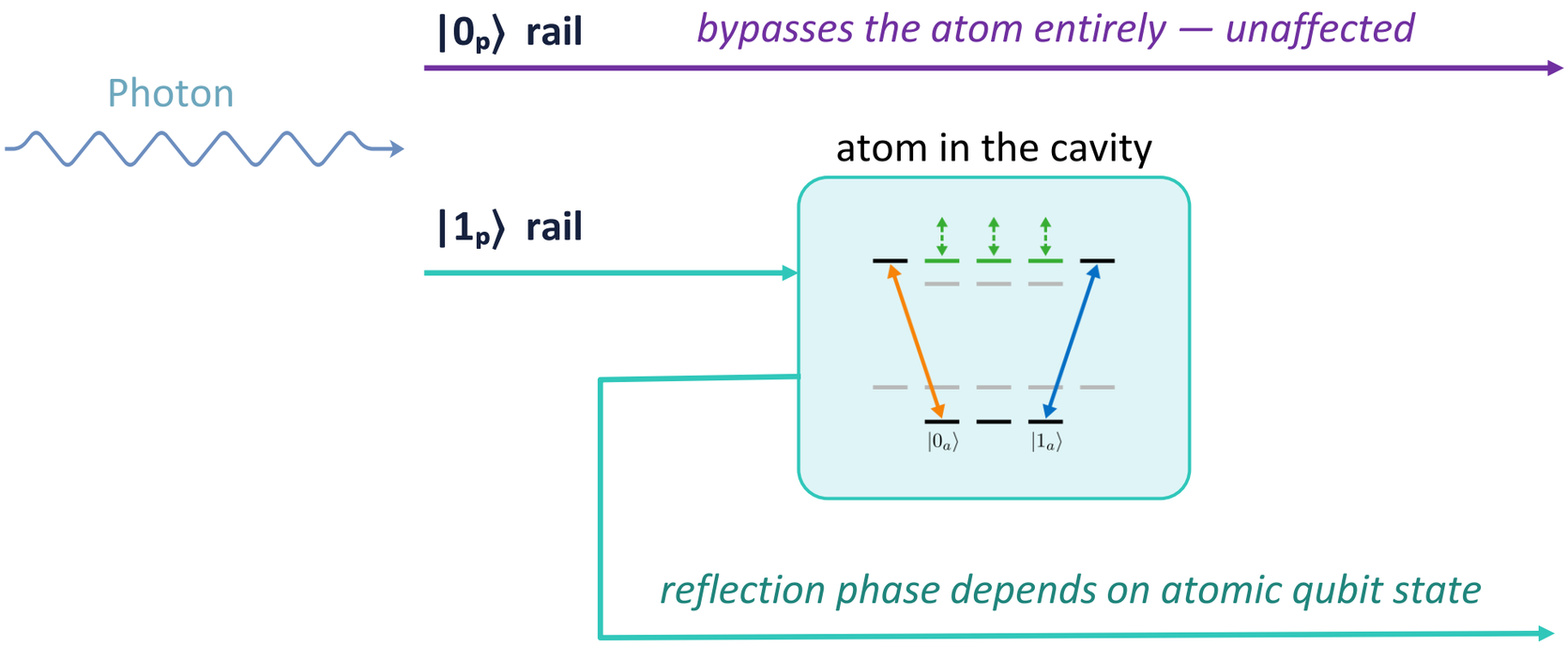
the same hardware, every role.

Coherent coupling at hundreds of MHz puts every operation on a ~10 ns timescale





The entangling (CZ) gate



$$|0_p\rangle|0_a\rangle \rightarrow |0_p\rangle|0_a\rangle$$

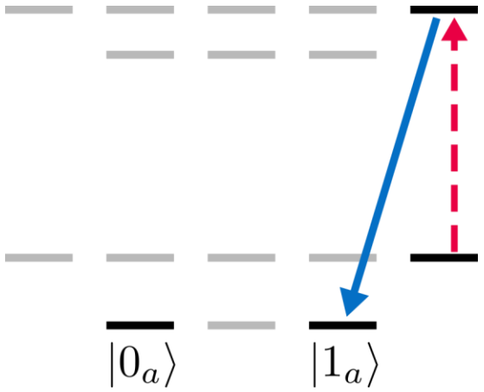
$$|0_p\rangle|1_a\rangle \rightarrow |0_p\rangle|1_a\rangle$$

$$|1_p\rangle|0_a\rangle \rightarrow |1_p\rangle|0_a\rangle$$

$$|1_p\rangle|1_a\rangle \rightarrow -|1_p\rangle|1_a\rangle$$

Core protocols

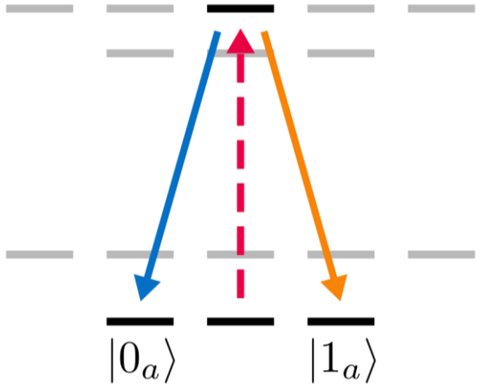
Single-Photon Generation



$$|F=2,m_F=2\rangle \xrightarrow{\text{vSTIRAP}} |F'=2,m_F=2\rangle$$
$$\rightarrow |F=1,m_F=0\rangle|1_p\rangle$$

Photon emitted into the cavity

Atomic State Preparation

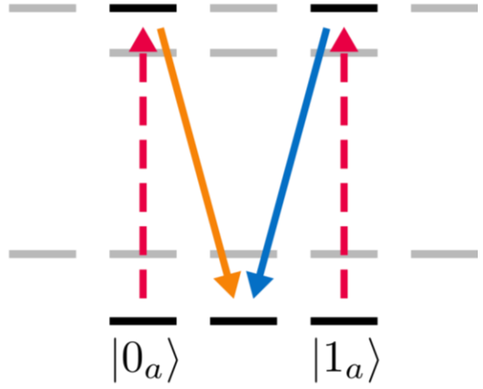


$$|F=1,m_F=0\rangle \xrightarrow{\text{vSTIRAP}} |F'=2,m_F=0\rangle$$
$$\rightarrow |0_a\rangle|1_p\rangle + |1_a\rangle|0_p\rangle$$
$$= |+_a\rangle|+_p\rangle - |-_a\rangle|-_p\rangle$$

Measure the photon:

atom is projected into $|\pm_a\rangle$, heralded!

Atomic State Measurement



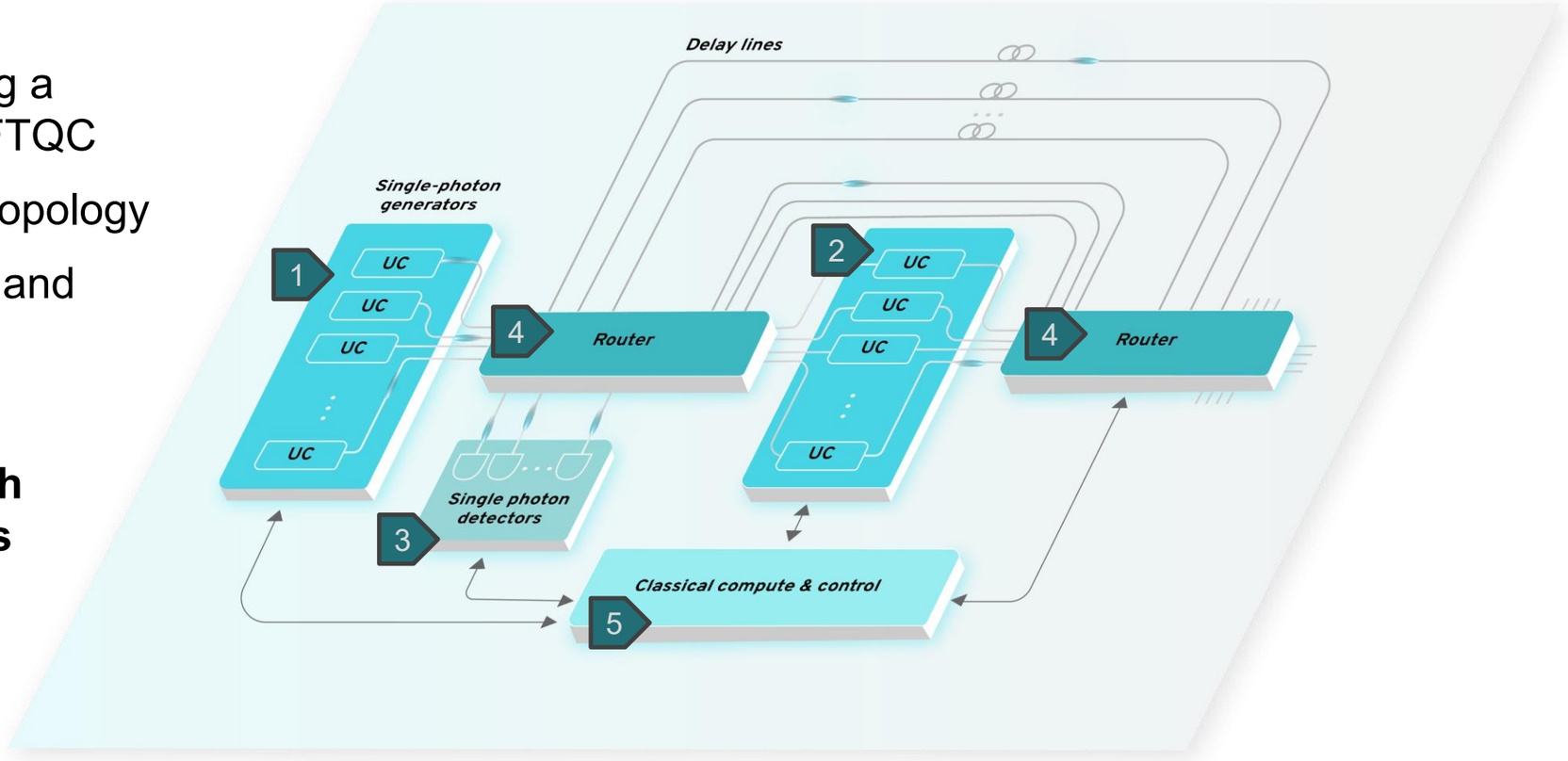
$$|\psi_a\rangle = \alpha|0_a\rangle + \beta|1_a\rangle$$
$$\xrightarrow{\text{vSTIRAP}} |F'=2,m_F=\pm 1\rangle$$
$$\rightarrow \alpha|0_a\rangle(\alpha|0_p\rangle + \beta|1_p\rangle)$$

Measure the photon; recycle the atom

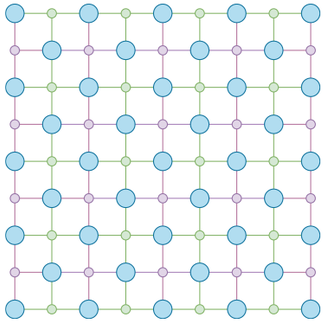
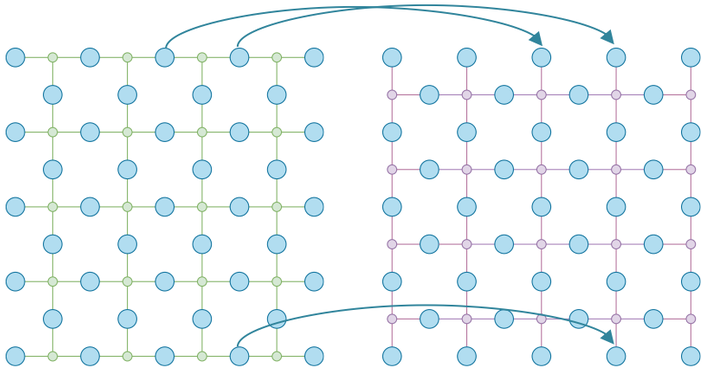
Putting the pieces together: the hardware stack

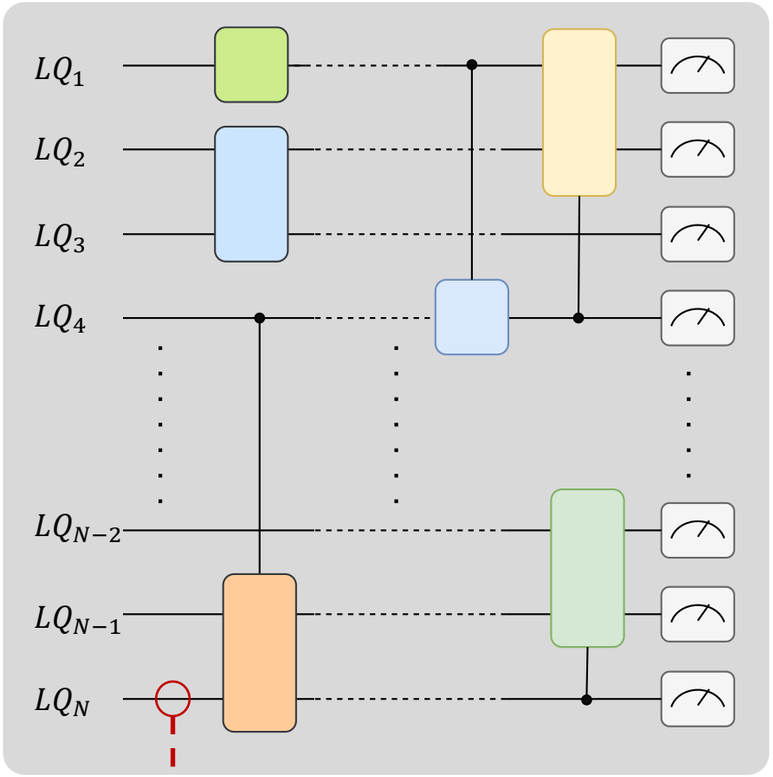
- On-demand photons are generated **(1)**
- And entangled **(2)** with atoms via CZ gates at unit cells (UCs),
- Then detected **(3)** by SNSPDs, building a cluster state for measurement-based FTQC
- Photon routing **(4)** creates the cluster topology
- Classical layer **(5)** controls the system and decodes errors in real time

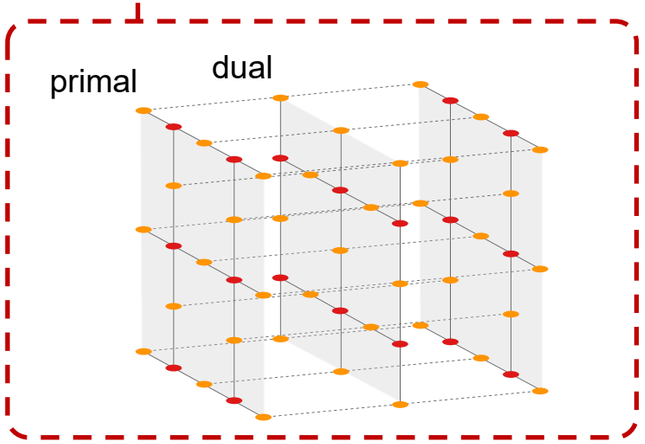
- **Scaling to any number of qubits, with simple replication of the subsystems**



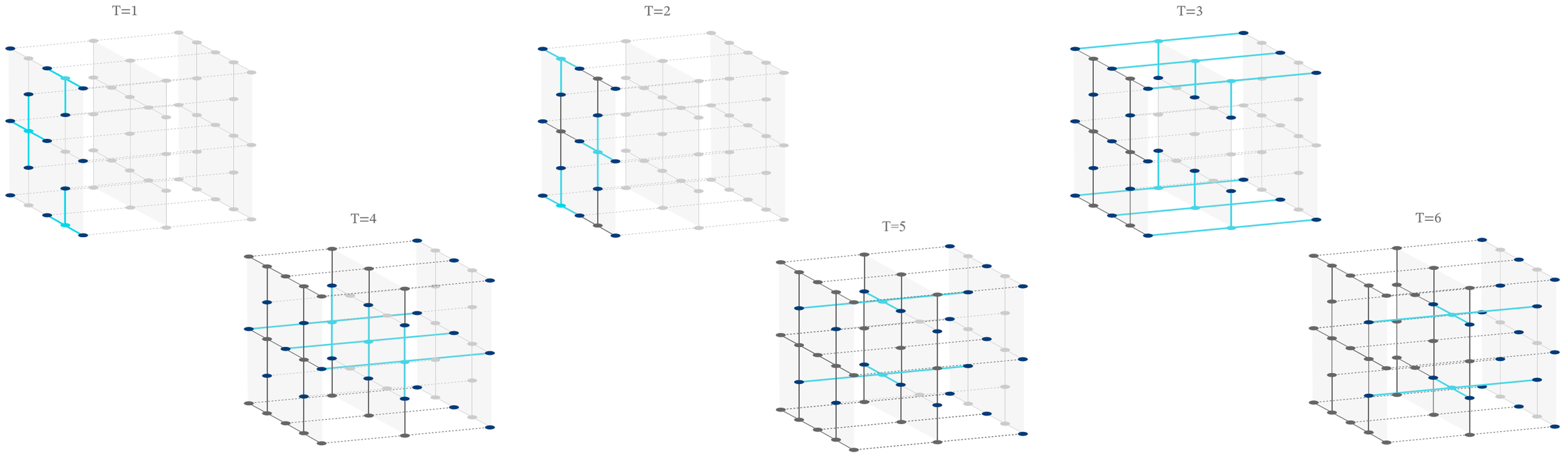
FTQC: what's under the hood?
(Take 2: CBQC vs MBQC)

CBQC: Logical qubit	MBQC: Logical qubit
	
Surface code	RHG lattice
Data qubits are stationary, carry quantum information	Data qubits are constantly measured out; quantum information is teleported forward
Each error correction (EC) cycle extracts X- and Z-syndrome information	Primal/dual layers contain X/Z syndrome-like measurements, which are then combined into parity checks. 2 layers ~ 1 EC cycle
For fault tolerance, ~10 EC cycles are needed to account for measurement errors	For fault tolerance, ~10 EC cycles are needed to account for measurement errors
Effectively, space-time syndrome data structures are fed into a decoder	Effectively, space-time syndrome data structures are fed into a decoder





RHG memory channel: bipartite generation scheme

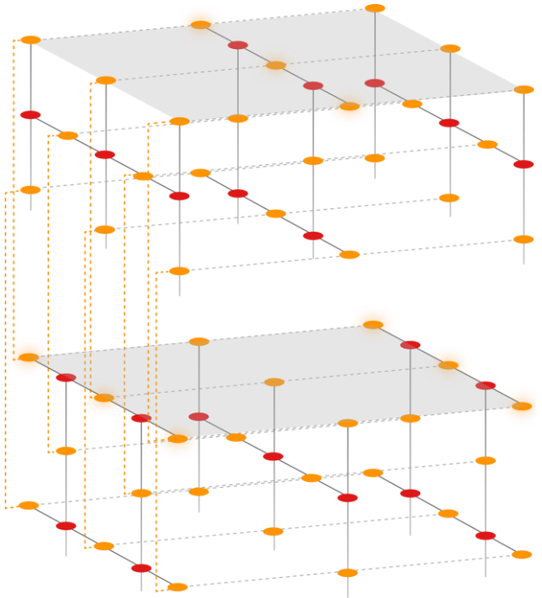


- Memory channel generation consists of photon cycles
- Each photon “sees” all its destination atoms almost simultaneously; each atom interacts with photons one by one.
- A single EC cycle corresponds to two measured layers
- With pulse width $\tau_p \simeq 25\,ns$, an EC cycle lasts 6 steps + change (for switching etc.) $\approx 200\,ns$

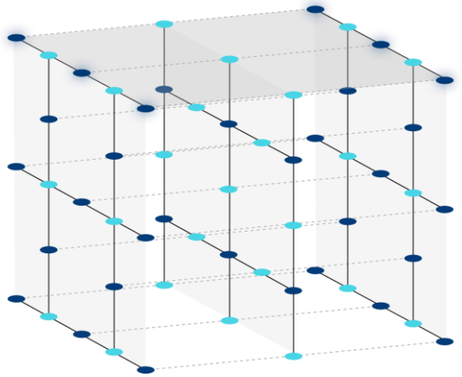
FTQC building blocks

- **Memory:** RHG lattice, adapted to compound photon-atom hardware, with a choice of generation schemes
- **Logic:** The full Clifford set — H, S, transversal CNOT — native to the RHG lattice
- **Universality:** Two independent paths to T-state preparation, based on code teleportation and magic state cultivation

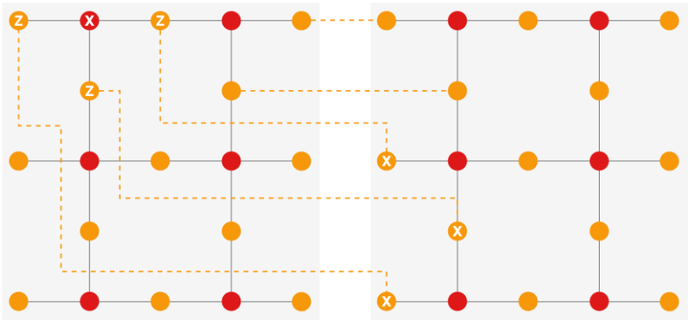
CNOT gate



Memory channel



Hadamard gate



Where our architectural edge compounds

- Faster EC cycle accelerates every logical layer and every factory cycle.
- Nonlocal photonic links avoid geometry-limited delivery of magic states and oracle data.
- Transversal Cliffords shrink the scaffolding around computation.

	Superconducting (Surface code)	Neutral Atoms (qLDPC)	Purely photonic (FBQC – topological code)	Quantum Source (RHG lattice)
Error-correction cycle	1 μs	~1 ms	1 μs – 10+ μs	200 ns
Routing overhead	Lattice surgery & SWAP chains	Atom rearrangement & Logical teleportation	Lattice surgery & interleaving	Minimal overhead (transversal)
Code selection	Limited by nearest-neighbor connectivity	Driven by # of qubits	Minimizing photon loss	Minimizing photon loss

Projected performance

Preliminary estimates — deeper resource analysis in progress

Benchmarks		Superconducting (Surface code)	Neutral Atoms (qLDPC)	Purely photonic (FBQC – topological code)	Quantum Source (RHG lattice)
RSA-2048	Runtime optimized	20Mq / 8 hours	100kq / 100 days	4800 RSG / 1.8 hours	8Mq / 1.5 hours
	HW footprint optimized	1Mq / 5 Days	10kq / 120 years	480 RSG / 18 hours	0.5Mq / 30 hours
FeMoco		4Mq / 4 days	?	Shows value of parallel factories	0.7-1.3Mq / 3-10 hours

- Gidney and Ekera, Quantum 5, 433 (2021).
- Gidney, arXiv:2505.15917 (2025).
- Cain et al, arXiv:2603.28627 (2026).
- Litinski and Nickerson,arXiv:2211.15465 (2022).
- Lee et al., PRX Quantum 2, 030305 (2021)
- Kim et al., Phys. Rev. Research 4, 023019 (2021).

Summary

- **Scale needs numbers and connectivity — we build for both**
  Deterministic photon–atom entanglement removes the resource tax every other approach pays somewhere.
- **The unit cell is a one-stop shop**
  One atom, one cavity, and every core operation — generation, entangling, prep, and readout — happens there.
- **This is a genuinely compound architecture, not atoms bolted onto photonics**
  Photons carry connectivity and scale, atoms provide determinism, and both carry quantum information
- **The FT building blocks are designed and simulated**
- **The edge compounds, it doesn't just add up**
  Faster EC cycles, nonlocal magic-state delivery, and lighter Clifford scaffolding